NOVEL-TO-MOVIE AI

Product Requirements Document (PRD)

Product vision, users, scope, outcomes and roadmap

Project	AI-powered multimodal story-to-film platform
Target event	Google AI Builder Cup 2026 hackathon
Document	PRD • NOVEL-TO-MOVIE-AI
Version	1.0 — Draft for implementation
Date	12 October 2026

Working principle

Turn a user-owned short novel into a coherent animated film through story understanding, screenplay planning, character and scene design, short-form video generation, audio, and automated editing.

Scope note: The 15/30/45/60-minute options are target runtimes for a future full pipeline. A hackathon MVP should prove the end-to-end workflow with a short sequence or trailer before attempting feature-length generation.

1. Executive summary

Novel-to-Movie AI is a web application that converts a user-provided short novel or story into an animated film. The system extracts the story's plot, characters, settings, tone and key events; creates a screenplay and scene plan; generates visual assets and short video clips; creates or coordinates dialogue, narration and sound; then assembles the clips into a playable, downloadable movie.

The product is designed around a human-reviewable production pipeline. Users can inspect the screenplay, approve character designs, regenerate individual scenes and review the final cut rather than trusting a single opaque generation step.

2. Problem statement

Turning prose into animation currently requires writing, storyboarding, character design, visual production, voice work, editing and technical skills. Existing generative tools can create short clips, but a long story needs orchestration, consistent characters, continuity management and editing across many generations.

The product opportunity is to make this process accessible while being honest about the cost, latency and quality constraints of long-form AI video.

3. Product vision and value proposition

- **Vision:** Let anyone turn an original story into a cinematic animated experience without needing a film-production team.
- **Value proposition:** One guided workflow from manuscript to structured screenplay, visual world, generated shots and assembled film.
- **Differentiator:** Story-aware planning and continuity management, not just isolated text-to-video prompts.
- **Primary demo outcome:** Upload a short story, approve characters and scenes, generate a short coherent film/trailer, and play or download the result.

4. Target users

Persona	Need	MVP value
Aspiring writer	Visualize a story without production expertise	A playable adaptation and editable scene plan
Indie creator / animator	Accelerate pre-production and concepting	Screenplay, character sheets, storyboards and generated shots
Educator / storyteller	Make narratives engaging for learners	Short narrated animated story
Hackathon judge	Understand the innovation quickly	Visible end-to-end workflow, quality controls and a working demo

5. Goals and success metrics

- A new user can upload a supported text file or paste a story and reach a generated preview through a guided flow.
- The system produces a structured story bible, screenplay and scene list before video generation.
- Users can approve or revise character descriptions and scene plans before spending video-generation credits.
- Every generated scene is traceable to its prompt, status, output asset and retry history.
- The demo completes successfully for a small, bounded story with clear progress and failure recovery.
- Measure: story-ingestion success rate; scene-plan approval rate; clip-generation success rate; character-consistency rating; final assembly success rate; average cost per demo; time to first preview.

6. Product scope

In scope for hackathon MVP

- Web UI with story upload/paste, genre/style choices, aspect ratio and short-demo runtime presets.
- Gemini-powered story analysis and structured JSON output for characters, locations, plot beats and screenplay.
- Character/setting reference images or character sheets, with user approval.
- Scene breakdown with per-scene prompts and estimated duration.
- Video generation through an available Google video-generation API/model, where enabled for the user's account.
- Basic narration/dialogue/audio handling and automated clip assembly.
- Job status, progress, retry for failed scenes, preview player and MP4 download.

Out of scope for the initial MVP

- Guaranteed production of polished 15–60-minute films on demand.
- Perfect lip-sync, identical character appearance in every shot, or studio-grade editing.
- Multi-user collaboration, marketplace, payments, mobile apps, advanced timeline editor or 3D rigging.
- Training a foundation model or hosting a custom video model.

7. Runtime strategy and feasibility

The UI may show intended runtime presets of 15, 30, 45 and 60 minutes, but these should be labelled as planned/advanced targets until the pipeline has been validated at scale. For the hackathon, provide a short preview/trailer target (for example, 30–90 seconds) and a scene-count cap.

Google's current Veo API documentation describes short clips of 4, 6 or 8 seconds per generation, depending on model/configuration. A 15-minute film is 900 seconds: even at 8 seconds per clip, it requires at least 113 clips before retries, transitions or discarded outputs. A 60-minute film requires at least 450 such clips. Real production would need more shots and would incur substantial latency and usage cost. Verify current model availability, quotas, pricing, supported regions and account access before committing to a target runtime.

Recommended product behaviour: let users choose a target runtime, calculate a rough shot/clip budget, display an estimate, and only enable long-form production when budget, quotas and pipeline reliability permit it.

8. Core user journey

Step	User experience	System outcome
1. Start project	Paste text or upload TXT/DOCX/PDF within limits	Validate file and extract text
2. Configure	Choose genre, animation style, language, aspect ratio and preview length	Persist project settings
3. Understand story	Review synopsis, characters, locations and plot beats	Gemini returns schema-constrained story data
4. Approve screenplay	Review and edit scene list	Lock approved screenplay version
5. Design world	Approve character sheets and visual style	Store reference assets and continuity notes
6. Generate	Start production and monitor progress	Queue scene jobs, call video model and track assets
7. Review/edit	Preview clips, retry a failed scene or regenerate a shot	Update scene assets and production status
8. Export	Play and download final cut	Assemble video/audio and provide export status

9. Product requirements (high level)

- **Story understanding:** identify characters, relationships, settings, timeline, conflict, climax and resolution.
- **Adaptation controls:** preserve core plot and user-selected tone; clearly label any adaptation choices.
- **Continuity:** maintain a canonical character/setting bible and reuse reference images and stable descriptions.
- **Scene planning:** create scenes with purpose, location, characters, action, dialogue/narration, visual prompt and target duration.
- **Production orchestration:** run generation asynchronously, track statuses, retry transient failures and avoid duplicate jobs.
- **Editing/export:** order clips, normalize audio where feasible, add title cards/subtitles optionally, and export a playable file.
- **Transparency:** show generation progress, limitations, estimated usage and failures without promising a guaranteed finish time.
- **Safety and rights:** require users to confirm they have rights to submitted stories; apply provider safety controls; do not intentionally imitate a living artist's style.

10. Proposed technology direction

Layer	Recommended MVP choice	Reason
Frontend	React + TypeScript	Fast web workflow, status and preview UI
Backend API	Python + FastAPI	Upload, validation, project and job APIs
Story reasoning	Gemini API via google-genai SDK; structured JSON/schema	Story extraction, screenplay and scene prompts
Video generation	Google video-generation model available to the account (e.g., Veo via Gemini API)	Short animated clips; verify current API/model access
Image assets	Gemini image-capable model/API available to account	Character sheets, environments and reference assets
Orchestration	Simple state machine/background worker first; LangGraph only if branching warrants it	Avoid over-engineering the hackathon MVP
Storage	Local/object storage for assets; SQLite for prototype metadata	Simple persistence; object storage for larger media
Assembly	FFmpeg	Join clips, mix audio and produce MP4
Deployment	Dockerized API + web app; managed storage if deployed	Reproducible demo deployment

11. UX principles

- Progressive disclosure: show story, screenplay and assets in stages instead of one overwhelming form.
- Human approval before expensive generation: screenplay and character design are review checkpoints.
- Visible progress: queued, generating, needs review, failed, assembling and completed.
- Scene-level recovery: a failed scene should not force the user to regenerate the whole film.
- Honest estimates: separate target runtime from generated runtime and show incomplete/placeholder scenes clearly.

12. Risks and mitigations

Risk	Impact	Mitigation
Long-form generation is costly/slow	Demo may time out or exceed budget	Cap MVP runtime and scenes; show estimate before launch
Character drift between clips	Movie feels incoherent	Reference sheets, canonical descriptions, visual review and selective regeneration
Model/API quota or access issues	Video step blocked	Check access early; create a clearly labelled storyboard/animatic fallback for demo resilience
Generation failures	Incomplete export	Async jobs, retries, per-scene status and partial preview
Story fidelity problems	Plot changes unexpectedly	Approve synopsis and screenplay; retain source-to-scene traceability
Copyright or unsafe content	Legal/safety risk	User rights confirmation, safety filtering and clear content policy
Uncontrolled spending	Unexpected API bill	Per-project budget cap, max clip count, estimate/consent and usage logging

13. Delivery roadmap

Phase	Deliverable	Definition of done
0 — Feasibility	Test API access and one clip end-to-end	Known model ID, limits, cost and working video response
1 — Story pipeline	Upload → story bible → screenplay → scene list	Structured data validates and can be edited
2 — Visual consistency	Character sheets and scene reference assets	User can approve references before generation
3 — Clip production	Queue and generate a bounded set of scenes	Progress, retries and stored clip assets work
4 — Assembly	Join clips, add narration/audio and export	MP4 plays and downloads
5 — Polish/demo	Progress UI, error states, sample story and demo script	A judge can complete the core flow without developer intervention

14. Hackathon demo definition

Recommended live demo: a 1–2 page original story; generated synopsis and cast; a screenplay with approximately 5–8 scenes; approved character references; a small set of generated clips; assembled 30–90 second preview; and a scene-level retry. If full video generation is not reliable during the event, demonstrate the complete planning flow plus a pre-generated sample clip, clearly identifying which assets are pre-generated.

The judging narrative should emphasize how Gemini interprets narrative context and how the pipeline turns that understanding into coherent, controllable multimedia output—not simply the fact that it calls a video API.

15. Open decisions

- Which video-generation model is enabled for the API key and region, and what are the current pricing and quota limits?
- Will the first MVP support TXT only, or TXT plus PDF/DOCX?
- Will the demo be stylized animation (recommended for a coherent look) or photorealistic output?
- Will speech be generated natively with clips, or added later using a separate text-to-speech service?
- What hard cap on scenes, clips and per-project spend should be enforced?

16. Reference links

Google AI for Developers — Veo video generation: https://ai.google.dev/gemini-api/docs/veo

Google AI for Developers — Gemini API documentation: https://ai.google.dev/gemini-api/docs

API capabilities, model IDs, availability, safety restrictions, pricing and quotas can change; verify them during Phase 0.